$$|\psi\rangle\langle\psi|^2 \;=\; |\psi\rangle\langle\psi|.$$

- The *trace* of an operator (or any square matrix) is defined as the sum of its diagonal elements. Using the symbol *Tr* for trace, we can define the trace of an operator **L** to be

$$Tr\ \mathbf{L} = \sum_i \langle i|\mathbf{L}|i\rangle,$$

  which is just the sum of **L**'s diagonal matrix elements.

  The trace of a projection operator is 1. This follows from the fact that the trace of a Hermitian operator is the sum of its eigenvalues.[4]

- If we add all the projection operators for a basis system, we obtain the identity operator:

$$\sum_i |i\rangle\langle i| = I. \qquad (7.11)$$

Finally, here is a very important theorem about projection operators and expectation values. The expectation value of any observable **L** in state |*ψ*> is given by

$$\langle\psi|\mathbf{L}|\psi\rangle = Tr\ |\psi\rangle\langle\psi|\ \mathbf{L}. \qquad (7.12)$$

Here are the steps to prove it. Pick any basis |*i*>. Then, using the definition of *trace*, write